Quadratic Penalty and Augmented Lagrangian Methods

Chapters 12 to 14 explained how to test an already available constrained candidate. We now turn to computing such candidates. Both methods in this chapter replace an equality-constrained problem by a sequence of unconstrained problems, so that the algorithms from Chapters 8 to 11 can serve as inner solvers. The quadratic penalty (QP) method forces feasibility by increasing a penalty parameter; the augmented Lagrangian method (ALM) also learns the constraint multipliers and can often keep that parameter finite. Both are developed from the primal problem. Duality and algorithms designed for the dual problem will be introduced later.

Learning objectives

By the end of this chapter, you should be able to:

- define and interpret the components of a KKT violation;
- derive the subproblems used by QP and ALM;
- implement their outer iterations with warm-started unconstrained solvers;
- distinguish global, approximately global, and stationary local inner solves;
- explain the trade-off between feasibility, inner accuracy, and conditioning; and
- state the assumptions needed for global or local convergence guarantees.

15.1 KKT violations and a two-level iterative framework

15.1.1 KKT violations as stopping criteria

Before constructing an algorithm, we need a numerical way to decide how closely a returned point satisfies the first-order conditions. For the smooth problem from Equation (12.2), write

$$\boldsymbol{g} = (g_1, \dots, g_m)^{\mathsf{T}}, \quad \boldsymbol{h} = (h_1, \dots, h_p)^{\mathsf{T}}.$$

$$\begin{aligned} \boldsymbol{A}(\boldsymbol{x}) &= [\nabla g_1(\boldsymbol{x}), \dots, \nabla g_m(\boldsymbol{x})], \\ \boldsymbol{B}(\boldsymbol{x}) &= [\nabla h_1(\boldsymbol{x}), \dots, \nabla h_p(\boldsymbol{x})]. \end{aligned}$$

The dimensions of $\boldsymbol{A}$ and $\boldsymbol{B}$ are $n \times m$ and $n \times p$. The Lagrangian is $L(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{\nu}) = f(\boldsymbol{x}) + \boldsymbol{\lambda}^{\mathsf{T}} \boldsymbol{g}(\boldsymbol{x}) + \boldsymbol{\nu}^{\mathsf{T}} \boldsymbol{h}(\boldsymbol{x})$. Its KKT first-order conditions are stationarity, primal feasibility, dual feasibility, and complementary slackness; see Equations (12.7) to (12.10). We measure the four violations by

$$r_{\text{stat}}(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{\nu}) = \left\| \nabla f(\boldsymbol{x}) + \boldsymbol{A}(\boldsymbol{x}) \boldsymbol{\lambda} + \boldsymbol{B}(\boldsymbol{x}) \boldsymbol{\nu} \right\|_{\infty},$$

$$
\begin{aligned}
r_{\text{feas}}(\boldsymbol{x}) &= \max\{\|\boldsymbol{g}(\boldsymbol{x})\|_\infty, \|[\boldsymbol{h}(\boldsymbol{x})]_+\|_\infty\},\\
r_{\text{dual}}(\boldsymbol{\nu}) &= \|[-\boldsymbol{\nu}]_+\|_\infty,\\
r_{\text{comp}}(\boldsymbol{x},\boldsymbol{\nu}) &= \|\boldsymbol{\nu}\odot\boldsymbol{h}(\boldsymbol{x})\|_\infty.
\end{aligned}
$$

Positive parts and the product $\odot$ are componentwise, i.e., $\mathbf{a}\odot\mathbf{b} := (a_i b_i)_i$. One aggregate is

$$r_{\text{KKT}}(\boldsymbol{x},\boldsymbol{\lambda},\boldsymbol{\nu}) = \max\{r_{\text{stat}}, r_{\text{feas}}, r_{\text{dual}}, r_{\text{comp}}\}.$$

The infinity norm matches the numerical implementation; other fixed norms give equivalent zero-residual and limiting conditions. Empty constraint blocks have norm zero. A norm without a subscript in the mathematical arguments below means the Euclidean norm.

To interpret the aggregate, observe that all four components are nonnegative. Their maximum is zero exactly when each vanishes: stationarity holds, $\boldsymbol{g} = \mathbf{0}$, $\boldsymbol{h} \le \mathbf{0}$, $\boldsymbol{\nu} \ge \mathbf{0}$, and $\nu_j h_j = 0$ for every j. Thus $r_{\text{KKT}} = 0$ is precisely the KKT first-order system.

The optimality results from earlier chapters determine what this means. Under LICQ, a local minimizer satisfies KKT FONC and SONC, whereas KKT SOSC certifies a strict local minimizer (Theorems 12.10, 13.5 and 13.11). For canonical convex problems, KKT is sufficient for global optimality; LICQ at a global minimizer gives necessity (Theorem 14.1). Consequently a small first-order residual is the numerical target here, but it does not certify a local minimum in a general nonconvex problem. Even for a convex problem, a positive residual alone supplies no quantitative distance or objective-error bound without further assumptions.

When choosing a threshold, also account for scaling. Multiplying a constraint by a positive constant preserves its feasible set but changes the residual and its multiplier. Separately scaled tolerances can therefore be preferable to one unscaled threshold. The examples below use the stated constraint scaling and a common outer threshold ε_{out}.

15.1.2 A two-level iterative framework

Instead of handling the constrained problem directly, we will turn to solve a sequence of easier subproblems. An *outer iteration*, indexed by k, constructs the next subproblem by updating its parameters or variables. The subproblems are unconstrained in this chapter and may retain some constraints in later chapters.

An *inner iteration* approximately minimizes the current subproblem. Denote its returned point by $\boldsymbol{x}^{(k+1)}$, independently of the number of inner steps. An individual inner iterate can be written $\boldsymbol{x}^{(k,\ell)}$. Superscripts with parentheses always indicate iteration indices, not powers.

For an unconstrained subproblem, possible inner solvers include line-search steepest descent (SD), safeguarded Newton, and nonlinear conjugate gradient from Chapters 8 to 11. Warm-starting a new inner solve from the previous returned point can save work. But the outer convergence guarantee depends on what the inner solver actually achieves, not simply on which algorithm is chosen.

Throughout this and subsequent chapters, we denote the inner tolerance by $\varepsilon_{\text{in}}^{(k)}$. Each stopping criterion specifies the residual or objective-value error that this tolerance bounds. The implemented QP and ALM algorithms keep this tolerance fixed: $\varepsilon_{\text{in}}^{(k)} \equiv \varepsilon_{\text{in}}^{(0)} > 0$. The outer stopping test uses $\varepsilon_{\text{out}} > 0$, chosen larger than this fixed inner tolerance. Alternative accuracy schedules and update rules are discussed only in Section 15.7; they are not implemented.

15.2 Equality-constrained optimization

We first develop the two methods for equality constraints, so that the relation between an inner gradient and the multiplier can be seen directly. Let $\mathcal{D} \subseteq \mathbb{R}^n$ be open and consider

$$\min_{\boldsymbol{x}\in\mathcal{D}} f(\boldsymbol{x}) \qquad \text{subject to} \qquad \boldsymbol{g}(\boldsymbol{x}) = \boldsymbol{0}. \tag{15.1}$$

Unless specified otherwise, f and $g_1, \dots, g_m$ are twice continuously differentiable. For this problem,

$$L(\boldsymbol{x}, \boldsymbol{\lambda}) = f(\boldsymbol{x}) + \boldsymbol{\lambda}^{\mathsf{T}} \boldsymbol{g}(\boldsymbol{x}), \qquad \boldsymbol{H}_L(\boldsymbol{x}, \boldsymbol{\lambda}) = \boldsymbol{H}_f(\boldsymbol{x}) + \sum_i \lambda_i \boldsymbol{H}_{g_i}(\boldsymbol{x}).$$

At a feasible point, LICQ means that $\boldsymbol{A}(\boldsymbol{x})$ has full column rank. Equality multipliers have no sign restrictions and no complementary slackness equations. The two relevant residuals are therefore

$$\begin{aligned} r_{\text{stat}}(\boldsymbol{x}, \boldsymbol{\lambda}) &= \left\| \nabla f(\boldsymbol{x}) + \boldsymbol{A}(\boldsymbol{x})\boldsymbol{\lambda} \right\|_\infty , \\ r_{\text{feas}}(\boldsymbol{x}) &= \left\| \boldsymbol{g}(\boldsymbol{x}) \right\|_\infty , \qquad r_{\text{KKT}}(\boldsymbol{x}, \boldsymbol{\lambda}) = \max\{r_{\text{stat}}, r_{\text{feas}}\}. \end{aligned} \tag{15.2}$$

15.3 QP method

15.3.1 Quadratic penalty function and its landscape geometry

To remove the equality constraints from the feasible set of an inner problem, assign an objective cost to their violation. The squared norm $\|\boldsymbol{g}(\boldsymbol{x})\|^2$ is nonnegative and vanishes exactly at feasible points, which leads to the following model.

Definition 15.1 (Quadratic penalty)**.** For $\rho > 0$, define

$$P_\rho(\boldsymbol{x}) = f(\boldsymbol{x}) + \frac{\rho}{2} \left\| \boldsymbol{g}(\boldsymbol{x}) \right\|^2 . \tag{15.3}$$

Increasing ρ penalizes equality violations more strongly. QP is also called the simple penalty method or Courant penalty method.

We will first examine this construction on one problem, then derive its stationarity equations and analyze what happens as ρ grows.

Example 15.1 (A common computational test)**.** For the geometry, penalty-growth, and method-cost comparisons, use

$$f(\boldsymbol{x}) = \frac{1}{2}[(x_1 - 2)^2 + 4(x_2 - 1)^2], \qquad g(\boldsymbol{x}) = x_1 + x_2 - 1.$$

To obtain a reference constrained solution, write the KKT equations:

$$x_1 - 2 + \lambda = 0, \qquad 4(x_2 - 1) + \lambda = 0, \qquad x_1 + x_2 = 1.$$

The first two give $x_1 = 2 - \lambda$ and $x_2 = 1 - \lambda/4$. Substitution into the equality gives $2 - 1.25\lambda = 0$, hence

$$\boldsymbol{x}^* = (0.4, 0.6)^{\mathsf{T}}, \qquad \lambda^* = 1.6, \qquad f(\boldsymbol{x}^*) = 1.6.$$

Strong convexity and an affine equality make this the unique global minimizer.

To visualize the effect of the penalty, Figure 15.1 plots the QP models for increasing ρ. The contours become steeper across the feasible line, while their centers move toward the constrained solution. We next connect this geometric observation to the equations solved by the inner method.

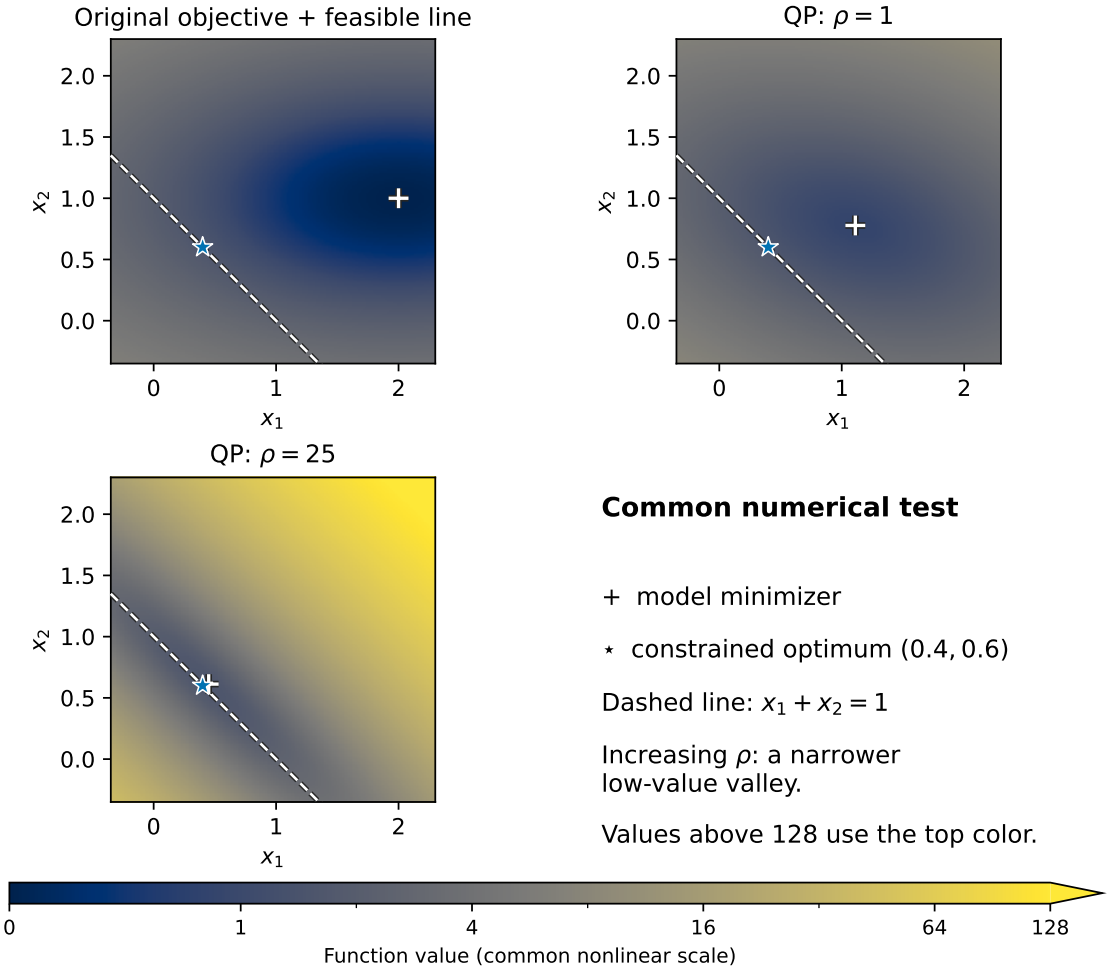


Figure 15.1: QP subproblem geometry for the common test: increasing ρ steepens the model across the feasible line and moves its inner minimizer toward the constrained minimizer.

15.3.2 From the inner gradient to KKT stationarity

Differentiate the squared constraint violation first. Since $\|\boldsymbol{g}(\boldsymbol{x})\|^2 = \sum_i g_i(\boldsymbol{x})^2$, the chain rule gives

$$\nabla\left(\tfrac{1}{2}\|\boldsymbol{g}(\boldsymbol{x})\|^2\right) = \sum_i g_i(\boldsymbol{x})\nabla g_i(\boldsymbol{x}) = \boldsymbol{A}(\boldsymbol{x})\boldsymbol{g}(\boldsymbol{x}).$$

Adding the objective gradient yields

$$\nabla P_\rho(\boldsymbol{x}) = \nabla f(\boldsymbol{x}) + \rho \boldsymbol{A}(\boldsymbol{x})\boldsymbol{g}(\boldsymbol{x}). \tag{15.4}$$

This has the form of Lagrangian stationarity if we set

$$\widehat{\boldsymbol{\lambda}}(\boldsymbol{x}, \rho) = \rho \boldsymbol{g}(\boldsymbol{x}). \tag{15.5}$$

With this estimate,

$$r_{\text{stat}}(\boldsymbol{x}, \widehat{\boldsymbol{\lambda}}) = \|\nabla P_\rho(\boldsymbol{x})\|_\infty .$$

Thus the inner gradient criterion controls exactly the stationarity component of the outer KKT test. It leaves feasibility to be checked separately.

For later use in Newton steps and conditioning estimates, differentiate once more. Each product $g_i \nabla g_i$ contributes $\nabla g_i \nabla g_i^{\mathsf{T}} + g_i \boldsymbol{H}_{g_i}$, so

$$\boldsymbol{H}_{P_\rho}(\boldsymbol{x}) = \boldsymbol{H}_f(\boldsymbol{x}) + \rho \sum_i [\nabla g_i(\boldsymbol{x}) \nabla g_i(\boldsymbol{x})^{\mathsf{T}} + g_i(\boldsymbol{x}) \boldsymbol{H}_{g_i}(\boldsymbol{x})]. \tag{15.6}$$

The coefficients of the constraint Hessians are precisely $\widehat{\lambda}_i$. Grouping them gives the useful decomposition

$$\boldsymbol{H}_{P_\rho}(\boldsymbol{x}) = \boldsymbol{H}_L(\boldsymbol{x}, \widehat{\boldsymbol{\lambda}}) + \rho \boldsymbol{A}(\boldsymbol{x}) \boldsymbol{A}(\boldsymbol{x})^{\mathsf{T}}.$$

15.3.3 What exact global inner solves guarantee

We now study an ideal sequence of globally minimized subproblems. The basic argument compares a penalty minimizer with a feasible point. If a global minimizer $\widehat{\boldsymbol{x}}$ of P_ρ is itself feasible, then every feasible $\boldsymbol{y}$ satisfies

$$f(\widehat{\boldsymbol{x}}) = P_\rho(\widehat{\boldsymbol{x}}) \le P_\rho(\boldsymbol{y}) = f(\boldsymbol{y}).$$

It is therefore already a global constrained minimizer. Usually the inner minimizer is infeasible; to understand that case, first compare the solutions at two different penalties.

Proposition 15.1 (Ideal penalty-path monotonicity)**.** *Let $0 < \rho_1 < \rho_2$, and let $\boldsymbol{x}_i$ globally minimize $P_{\rho_i}(\cdot)$. Then*

$$\|\boldsymbol{g}(\boldsymbol{x}_2)\| \le \|\boldsymbol{g}(\boldsymbol{x}_1)\|, \qquad f(\boldsymbol{x}_1) \le f(\boldsymbol{x}_2).$$

Proof. Put $v_i = \|\boldsymbol{g}(\boldsymbol{x}_i)\|^2$ and $f_i = f(\boldsymbol{x}_i)$. Global optimality gives

$$f_1 + \frac{\rho_1}{2} v_1 \le f_2 + \frac{\rho_1}{2} v_2, \qquad f_2 + \frac{\rho_2}{2} v_2 \le f_1 + \frac{\rho_2}{2} v_1.$$

Adding yields $(\rho_2 - \rho_1)(v_2 - v_1) \le 0$, hence $v_2 \le v_1$. The first inequality gives $f_1 - f_2 \le (\rho_1/2)(v_2 - v_1) \le 0$. □

The two-penalty comparison shows a monotone trade-off between objective value and squared violation. To identify a limiting point, compare every subproblem instead with one fixed feasible point.

Theorem 15.2 (Cluster points of globally solved penalty subproblems)**.** *Suppose $f, \boldsymbol{g}$ are continuous, the feasible set is nonempty, $\rho^{(k)} \to \infty$, and $\boldsymbol{x}^{(k+1)}$ globally minimizes $P_{\rho^{(k)}}(\cdot)$. Every cluster point of the returned points belonging to $\mathcal{D}$ globally minimizes Equation (15.1). If $f(\boldsymbol{x}^{(k+1)})$ is bounded below, their whole sequence of feasibility violations tends to zero.*

Proof. Fix any feasible $\boldsymbol{y}$. Comparison with it gives

$$f(\boldsymbol{x}^{(k+1)}) + \frac{\rho^{(k)}}{2} \left\| \boldsymbol{g}(\boldsymbol{x}^{(k+1)}) \right\|^2 \le f(\boldsymbol{y}). \tag{15.7}$$

If $\boldsymbol{x}^{(k_j+1)} \to \boldsymbol{x}^* \in \mathcal{D}$, continuity bounds the objective along this subsequence. Rearranging gives

$$\left\|\boldsymbol{g}(\boldsymbol{x}^{(k_j+1)})\right\|^2 \leq \frac{2[f(\boldsymbol{y}) - f(\boldsymbol{x}^{(k_j+1)})]}{\rho^{(k_j)}} \to 0.$$

Thus $\boldsymbol{g}(\boldsymbol{x}^*) = \boldsymbol{0}$. Nonnegativity of the penalty also gives $f(\boldsymbol{x}^{(k_j+1)}) \leq f(\boldsymbol{y})$; taking limits proves $f(\boldsymbol{x}^*) \leq f(\boldsymbol{y})$. Since $\boldsymbol{y}$ was arbitrary, global optimality follows. If $f(\boldsymbol{x}^{(k+1)}) \geq a$ for every k, the same estimate with $f(\boldsymbol{y}) - a$ proves whole-sequence feasibility. □

The theorem identifies interior cluster points when they exist. To ensure that such points exist, we can place the whole sequence in one compact sublevel set.

Corollary 15.3 (A compactness condition ensuring cluster points)**.** *Under the preceding assumptions, suppose for some feasible $\boldsymbol{y}$ and $\rho_0 > 0$ that $\{\boldsymbol{x} \in \mathcal{D} : P_{\rho_0}(\boldsymbol{x}) \leq f(\boldsymbol{y})\}$ is compact. If $\rho^{(k)} \geq \rho_0$, all returned points belong to this compact set; cluster points exist and are globally optimal.*

Proof. For every $\boldsymbol{x}$, $P_{\rho_0}(\boldsymbol{x}) \leq P_{\rho^{(k)}}(\boldsymbol{x})$. Hence

$$P_{\rho_0}(\boldsymbol{x}^{(k+1)}) \leq P_{\rho^{(k)}}(\boldsymbol{x}^{(k+1)}) \leq P_{\rho^{(k)}}(\boldsymbol{y}) = f(\boldsymbol{y}).$$

Compactness and Theorem 15.2 finish the proof. □

The global hypotheses are essential here. Existence of inner minimizers, bounded returned points, and interior cluster points must be checked rather than inferred from the update alone. Also, the monotonicity result concerns the Euclidean norm in the penalty. With several equalities it need not imply monotonicity of the recorded infinity norm; for the scalar equality in the common test the norms coincide. We now turn from this ideal analysis to the finite-accuracy algorithm used in the computations. Extensions to approximately global or stationary inner solutions appear in Theorems 15.8 and 15.9.

15.3.4 Outer algorithm and inner solvers

Use Equation (15.5) to evaluate stationarity after each inner solve. If the full KKT test still fails, increase the penalty by a prescribed factor and solve the next model from the returned point.

Algorithm: QP method

Choose $\boldsymbol{x}^{(0)}$, $\rho^{(0)} > 0$, growth factor $\gamma > 1$, a fixed inner tolerance $0 < \varepsilon_{\text{in}}^{(k)} \equiv \varepsilon_{\text{in}}^{(0)} < \varepsilon_{\text{out}}$, and outer KKT tolerance $\varepsilon_{\text{out}} > 0$. At iteration k:

1. Warm-start from $\boldsymbol{x}^{(k)}$ and approximately minimize $P_{\rho^{(k)}}(\cdot)$ until
$$\left\|\nabla P_{\rho^{(k)}}(\boldsymbol{x}^{(k+1)})\right\|_\infty \leq \varepsilon_{\text{in}}^{(k)}.$$
2. Set $\widehat{\boldsymbol{\lambda}}^{(k+1)} = \rho^{(k)}\boldsymbol{g}(\boldsymbol{x}^{(k+1)})$, and compute both equality residuals.
3. Stop if $r_{\text{KKT}}(\boldsymbol{x}^{(k+1)}, \widehat{\boldsymbol{\lambda}}^{(k+1)}) \leq \varepsilon_{\text{out}}$. Otherwise set $\rho^{(k+1)} = \gamma\rho^{(k)}$ and continue with the same inner tolerance.

Both implemented methods use warm-started SD or safeguarded Newton with Armijo backtracking. An inner solve has no iteration limit: it continues until its gradient criterion holds or reports numerical breakdown or stagnation. An unsuccessful inner solve stops the outer run.

15.3.5 The limitation of local inner solves

To understand what the inner gradient criterion can guarantee, we next examine a subproblem with an infeasible local minimizer. This example uses fully solved local subproblems and therefore applies even when the fixed inner tolerance is met exactly. Deliberately truncating an inner solve raises a separate issue, discussed in Section 15.7.5.

Example 15.2 (An infeasible strict local minimum of every subproblem)**.** Take $f(x) = (x-1)^2$ and $g(x) = x^3 - 3x + 3$. To locate the feasible set, differentiate g. Its critical points $-1, 1$ have values $5, 1$; hence the increasing left branch crosses zero once and the other branches remain positive. The unique feasible point is $x^* \approx -2.1038034$. At this root $g'(x^*) = 3[(x^*)^2 - 1] \neq 0$, so LICQ holds. But at $x = 1$,

$$g(1) = 1, \qquad P'_\rho(1) = 0, \qquad P''_\rho(1) = 2 + 6\rho > 0.$$

This is an infeasible strict local minimum for every $\rho > 0$. A local solver warm-started there can stay there forever, even with exact local inner solves. Globally solved penalty problems instead approach x^*, by the global theorem and compactness of their sublevel sets. Thus exact local and exact global inner solves are not interchangeable.

15.3.6 Why high feasibility accuracy can require a large penalty

To quantify the penalty needed for feasibility, solve the common quadratic subproblem exactly. Write $\widehat{\lambda} = \rho(x_1 + x_2 - 1)$. The two stationarity equations then give

$$x_1 = 2 - \widehat{\lambda}, \qquad x_2 = 1 - \widehat{\lambda}/4.$$

Substitute these expressions into the multiplier estimate:

$$\widehat{\lambda} = \rho(2 - 1.25\widehat{\lambda}), \qquad (1 + 1.25\rho)\widehat{\lambda} = 2\rho.$$

Consequently

$$\widehat{\lambda}(\rho) = \frac{2\rho}{1 + 1.25\rho}, \qquad \boldsymbol{x}(\rho) = \begin{pmatrix} 2 - \widehat{\lambda}(\rho) \\ 1 - \widehat{\lambda}(\rho)/4 \end{pmatrix}, \qquad g(\boldsymbol{x}(\rho)) = \frac{2}{1 + 1.25\rho}.$$

The last expression is positive for every finite penalty and decreases only at order $1/\rho$. For example, feasibility below ε_{out} requires $\rho \geq (2/\varepsilon_{\text{out}} - 1)/1.25$ when that lower bound is positive. This limitation remains even with exact inner minimization. To state the corresponding general issue, introduce exactness.

Definition 15.2 (Local exactness at a constrained solution)**.** A penalty family is locally exact at $\boldsymbol{x}^*$ if, for every sufficiently large finite penalty parameter, $\boldsymbol{x}^*$ is a local minimizer of the penalized problem.

Global exactness requires coincidence of the corresponding global solution sets. To determine whether QP can be locally exact, evaluate its gradient at a constrained KKT point.

Proposition 15.4 (A nonzero regular multiplier prevents finite exactness)**.** *Let $(\boldsymbol{x}^*, \boldsymbol{\lambda}^*)$ be a feasible KKT pair with LICQ and $\boldsymbol{\lambda}^* \neq \mathbf{0}$. Then $\boldsymbol{x}^*$ is not stationary, and hence not a local minimizer, for $P_\rho(\cdot)$ at any finite ρ. If $\boldsymbol{x}^{(k+1)} \to \boldsymbol{x}^*$ and $\widehat{\boldsymbol{\lambda}}^{(k+1)} \to \boldsymbol{\lambda}^* \neq \mathbf{0}$, no bounded subsequence of corresponding penalties is possible.*

Proof. At feasibility, $\nabla P_\rho(\boldsymbol{x}^*) = \nabla f(\boldsymbol{x}^*) = -\boldsymbol{A}(\boldsymbol{x}^*)\boldsymbol{\lambda}^* \neq \mathbf{0}$ by full column rank. The unconstrained first-order necessary condition excludes local optimality. A bounded penalty subsequence, together with $\boldsymbol{g}(\boldsymbol{x}^{(k+1)}) \to \mathbf{0}$, would give $\widehat{\boldsymbol{\lambda}}^{(k+1)} = \rho^{(k)}\boldsymbol{g}(\boldsymbol{x}^{(k+1)}) \to \mathbf{0}$, contradicting the nonzero multiplier limit. □

15.3.7 Why a large penalty can make inner solves difficult

The preceding calculation calls for larger penalties as the feasibility target tightens. We next examine the cost of solving those subproblems. In the decomposition $\boldsymbol{H}_{P_\rho} = \boldsymbol{H}_L + \rho\boldsymbol{A}\boldsymbol{A}^\mathsf{T}$, the added term acts on normal directions and vanishes on tangent directions. Comparing these two types of directions gives a condition-number estimate.

Proposition 15.5 (Normal–tangent curvature separation)**.** *Suppose $\rho^{(k)} \to \infty$ and penalty points tend to a point where $\boldsymbol{A}$ has full column rank $1 \leq m < n$. Suppose their $\boldsymbol{H}_L(\boldsymbol{x}^{(k+1)}, \widehat{\boldsymbol{\lambda}}^{(k+1)})$ have uniformly bounded norm and their penalty Hessians are positive definite. Then some constants $\sigma, M > 0$ give, for large k,*

$$\lambda_{\max}(\boldsymbol{H}_{P_\rho}) \geq \rho^{(k)}\sigma^2 - M, \qquad \lambda_{\min}(\boldsymbol{H}_{P_\rho}) \leq M.$$

Consequently the spectral condition numbers grow at least on the scale of $\rho^{(k)}$.

Proof. Continuity and full column rank give a uniform positive lower bound σ on the smallest singular value of $\boldsymbol{A}$. For a unit normal vector $\boldsymbol{y} \in \mathrm{Range}(\boldsymbol{A})$, $\left\|\boldsymbol{A}^\mathsf{T}\boldsymbol{y}\right\| \geq \sigma$, and hence

$$\boldsymbol{y}^\mathsf{T}\boldsymbol{H}_{P_\rho}\boldsymbol{y} = \boldsymbol{y}^\mathsf{T}\boldsymbol{H}_L\boldsymbol{y} + \rho^{(k)}\left\|\boldsymbol{A}^\mathsf{T}\boldsymbol{y}\right\|^2 \geq -M + \rho^{(k)}\sigma^2.$$

This bounds the largest eigenvalue from below. Since $m < n$, a unit tangent vector $\boldsymbol{z} \in \mathrm{Null}(\boldsymbol{A}^\mathsf{T})$ exists, and $\boldsymbol{z}^\mathsf{T}\boldsymbol{H}_{P_\rho}\boldsymbol{z} = \boldsymbol{z}^\mathsf{T}\boldsymbol{H}_L\boldsymbol{z} \leq M$. This bounds the smallest eigenvalue from above. Positive definiteness permits division to obtain the condition-number lower bound. □

For the common test,

$$\boldsymbol{H}_{P_\rho} = \begin{pmatrix} 1+\rho & \rho \\ \rho & 4+\rho \end{pmatrix}, \qquad \lambda_\pm(\boldsymbol{H}_{P_\rho}) = \frac{5 + 2\rho \pm \sqrt{9 + 4\rho^2}}{2}.$$

Therefore $\lambda_- \to 2.5$, $\lambda_+ \sim 2\rho$, and $\kappa(\boldsymbol{H}_{P_\rho}) \sim 0.8\rho$. The objective remains strictly convex, but SD becomes increasingly sensitive to elongated contours.

Warning

Large curvature is not singularity: conditioning concerns a ratio of scales. Nor does $\rho \to \infty$ invariably cause a diverging condition number. When $m = n$, there are no tangent directions. For example, $f(\boldsymbol{x}) = \|\boldsymbol{x}\|^2/2$, $\boldsymbol{g}(\boldsymbol{x}) = \boldsymbol{x}$ gives $\boldsymbol{H}_{P_\rho} = (1+\rho)\boldsymbol{I}$,

whose condition number is one.

To measure how the fixed growth factor affects total work, run QP four times with different choices of γ, holding each choice fixed throughout its run. Keep the SD inner gradient tolerance at $\varepsilon_{\text{in}}^{(k)} \equiv 10^{-4}$, request $r_{\text{KKT}} \le 10^{-3}$, and use the common quadratic test with $\boldsymbol{x}^{(0)} = (2,1)^{\mathsf{T}}$, $\rho^{(0)} = 1$, and warm starts. All four growth-factor runs below reach the requested KKT tolerance without an inner iteration limit.

Growth factor γ	Outer solves	Total SD steps	Objective evaluations
2	12	2673	34042
5	6	4998	67985
10	5	9755	147284
20	4	13841	215768

Objective evaluations include Armijo trials and the solver's objective checks. For $\gamma = 2$, increasing ρ from 16 to 2048 increases the SD steps for one subproblem from 25 to 1323, although the requested inner accuracy is unchanged; see Figure 15.2. Individual counts need not increase at every stage. This is an observed cost trade-off for this problem and subsolver, not a claim that one growth factor is always best.

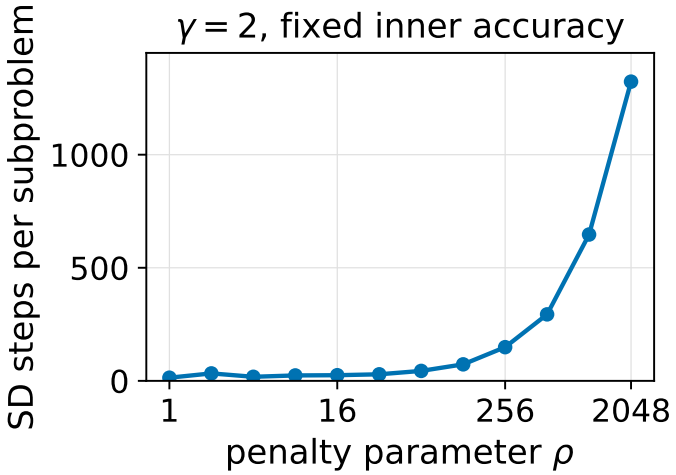


Figure 15.2: QP with fixed inner accuracy: larger penalties make SD inner solves more expensive, and faster penalty growth can increase total SD steps and objective evaluations despite reducing the outer-solve count.

15.4 ALM

15.4.1 Retaining a multiplier estimate and the subproblem

We now seek a model that can approach a nonzero multiplier without requiring an unbounded penalty. The QP identity $\widehat{\boldsymbol{\lambda}} = \rho \boldsymbol{g}$ suggests retaining the multiplier information from one subproblem to the next. Add a current estimate through the ordinary Lagrangian, then keep the quadratic term to penalize constraint violation.

Definition 15.3 (Equality augmented Lagrangian)**.** For $\rho > 0$, define

$$L_A(\boldsymbol{x}, \boldsymbol{\lambda}; \rho) = f(\boldsymbol{x}) + \boldsymbol{\lambda}^\mathsf{T} \boldsymbol{g}(\boldsymbol{x}) + \frac{\rho}{2} \|\boldsymbol{g}(\boldsymbol{x})\|^2 . \tag{15.8}$$

For a current multiplier $\boldsymbol{\lambda}^{(k)}$, the inner problem minimizes $L_A(\cdot, \boldsymbol{\lambda}^{(k)}; \rho)$ without equality constraints.

To identify the multiplier associated with a returned inner point, differentiate the linear multiplier term and use Equation (15.4) for the penalty term:

$$\nabla_{\boldsymbol{x}} L_A(\boldsymbol{x}, \boldsymbol{\lambda}; \rho) = \nabla f(\boldsymbol{x}) + \boldsymbol{A}(\boldsymbol{x})[\boldsymbol{\lambda} + \rho \boldsymbol{g}(\boldsymbol{x})]. \tag{15.9}$$

The bracketed coefficient is the estimate that makes the inner gradient equal to the Lagrangian gradient. At $\boldsymbol{x}^{(k+1)}$, we therefore define

$$\boldsymbol{\lambda}^{(k+1)} = \boldsymbol{\lambda}^{(k)} + \rho \boldsymbol{g}(\boldsymbol{x}^{(k+1)}). \tag{15.10}$$

It follows immediately that

$$r_{\text{stat}}(\boldsymbol{x}^{(k+1)}, \boldsymbol{\lambda}^{(k+1)}) = \left\| \nabla_{\boldsymbol{x}} L_A(\boldsymbol{x}^{(k+1)}, \boldsymbol{\lambda}^{(k)}; \rho) \right\|_\infty .$$

The penalty term now supplies a correction to the retained estimate. This derivation applies at every returned inner point, so the classical update is unconditional; ρ stays fixed in the algorithm below.

Finally, differentiate again and group the coefficients of $\boldsymbol{H}_{g_i}$ as in Equation (15.6). This gives

$$\boldsymbol{H}_{L_A}(\boldsymbol{x}, \boldsymbol{\lambda}; \rho) = \boldsymbol{H}_L(\boldsymbol{x}, \boldsymbol{\lambda} + \rho \boldsymbol{g}(\boldsymbol{x})) + \rho \boldsymbol{A}(\boldsymbol{x}) \boldsymbol{A}(\boldsymbol{x})^\mathsf{T}. \tag{15.11}$$

15.4.2 Augmented Lagrangian landscapes and finite-parameter exactness

Example 15.3 (Finite-parameter contours and an exact multiplier recurrence)**.** We can compute the multiplier correction explicitly for the common test. For a current estimate λ, write $\lambda^+ = \lambda + \rho g(\boldsymbol{x})$. The inner stationarity equations again give $x_1 = 2 - \lambda^+$ and $x_2 = 1 - \lambda^+/4$, so

$$g(\boldsymbol{x}) = 2 - 1.25\lambda^+, \qquad \lambda^+ = \lambda + \rho(2 - 1.25\lambda^+).$$

Solving this scalar equation gives

$$\lambda^+ = \frac{\lambda + 2\rho}{1 + 1.25\rho}, \qquad g(\boldsymbol{x}(\lambda, \rho)) = \frac{2 - 1.25\lambda}{1 + 1.25\rho}.$$

To compare the new estimate with $\lambda^* = 1.6$, subtract this fixed point from both sides:

$$\lambda^+ - 1.6 = \frac{\lambda - 1.6}{1 + 1.25\rho}, \qquad \boldsymbol{x}(\lambda, \rho) = \begin{pmatrix} 2 - \lambda^+ \\ 1 - \lambda^+/4 \end{pmatrix}.$$

At $\rho = 10$, the multiplier error contracts by $1/13.5$ per outer solve. Once $\lambda = 1.6$, the inner minimizer is exactly $\boldsymbol{x}^*$; no increase of ρ is needed for this example. Figure 15.3 shows the corresponding movement of the augmented contour center.

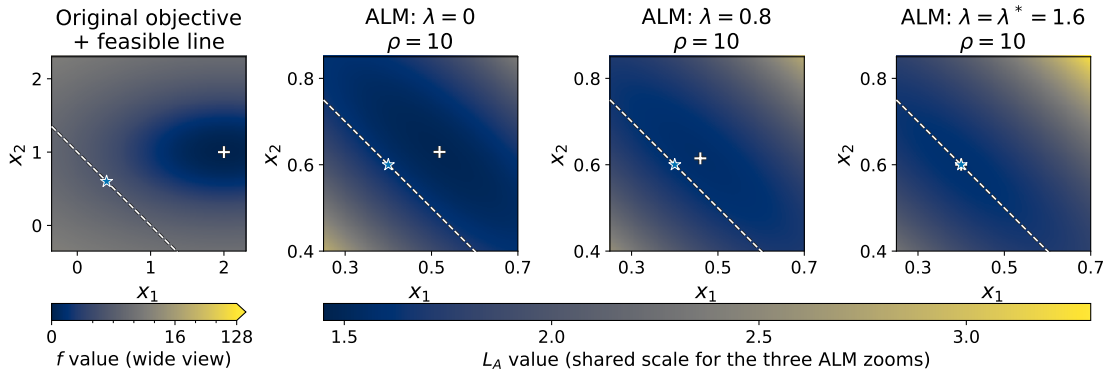


Figure 15.3: At fixed $\rho = 10$, the augmented landscapes for $\lambda = 0, 0.8, 1.6$ share one local window and one absolute-value scale. Their inner minimizers approach $\boldsymbol{x}^*$ as the multiplier estimate approaches $\lambda^* = 1.6$; at the correct multiplier, $\boldsymbol{x}^*$ is the global inner minimizer.

To see what can change when the objective is nonconvex, fix the correct multiplier and examine the curvature away from the feasible set. Consider

$$\min_{\boldsymbol{x}\in\mathbb{R}^2} \quad f(\boldsymbol{x}) = -\frac{1}{2}x_1^2 + \frac{1}{2}x_2^2, \qquad \text{subject to} \qquad g(\boldsymbol{x}) = x_1 = 0. \tag{15.12}$$

On the feasible line, the objective reduces to $x_2^2/2$, so the unique solution is $\boldsymbol{x}^* = \boldsymbol{0}$. Stationarity of $L(\boldsymbol{x}, \lambda) = f(\boldsymbol{x}) + \lambda x_1$ gives $\lambda^* = 0$. The constraint gradient $(1, 0)^\mathsf{T}$ is nonzero, so LICQ holds. A tangent direction satisfies $d_1 = 0$, and hence

$$\boldsymbol{d}^\mathsf{T}\boldsymbol{H}_L(\boldsymbol{0}, 0)\boldsymbol{d} = -d_1^2 + d_2^2 = d_2^2 > 0 \quad \text{whenever } \boldsymbol{d} \neq \boldsymbol{0} \text{ and } d_1 = 0.$$

Thus SOSC holds on a nontrivial tangent space. In contrast, the normal x_1-direction, along which $x_2 = 0$, has curvature -1. An unconstrained inner minimization can move in this direction, so we must examine whether the penalty offsets that negative curvature.

At the correct multiplier, adding the penalty gives

$$L_A(\boldsymbol{x}, 0; \rho) = \frac{\rho - 1}{2}x_1^2 + \frac{1}{2}x_2^2, \qquad \boldsymbol{H}_{L_A} = \begin{pmatrix} \rho - 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

The tangent curvature stays equal to 1, while the normal curvature increases from -1 to $\rho - 1$. This yields the three landscapes in Figure 15.4:

- If $0 < \rho < 1$, the normal curvature is still negative. The origin is a saddle point of L_A, whose values tend to $-\infty$ along the x_1-axis.
- If $\rho = 1$, the normal curvature is zero. Every point $(x_1, 0)$ globally minimizes L_A, so the origin is a non-strict global minimizer in a flat valley. All the other minimizers are infeasible for the constrained problem.
- If $\rho > 1$, both curvatures are positive. The origin is the unique global inner minimizer, as well as the constrained solution.

For later use in the outer iteration, also allow a current multiplier $\lambda \neq \lambda^*$. The additional term is linear, so for $\rho > 1$ the inner objective remains strongly convex:

$$L_A(\boldsymbol{x}, \lambda; \rho) = \frac{\rho - 1}{2}x_1^2 + \lambda x_1 + \frac{1}{2}x_2^2.$$

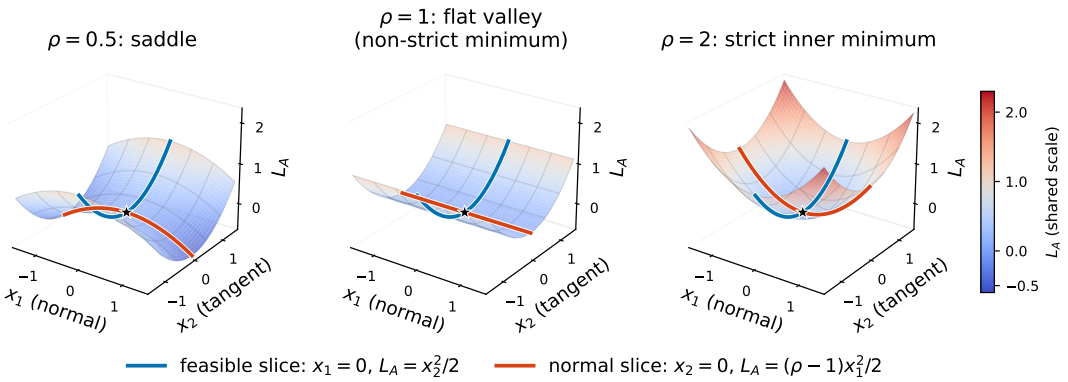


Figure 15.4: The augmented landscape for Equation (15.12) at the correct multiplier $\lambda^* = 0$: $\rho = 0.5$ leaves a saddle, $\rho = 1$ gives a flat valley of global minimizers, and $\rho = 2$ gives a unique strict minimizer at the constrained solution.

Setting its partial derivatives to zero gives

$$(\rho - 1)x_1 + \lambda = 0, \qquad x_2 = 0 \quad \Longrightarrow \quad \boldsymbol{x}(\lambda; \rho) = \left(-\frac{\lambda}{\rho - 1}, 0\right)^{\mathsf{T}}. \tag{15.13}$$

Thus $\rho > 1$ ensures a unique global inner solution for every multiplier. At the boundary $\rho = 1$, a nonzero λ adds a slope to the flat x_1-direction and makes the inner objective unbounded below.

We can now identify the same curvature mechanism at a general feasible KKT pair. The augmented gradient at the correct multiplier vanishes. By Equation (15.11), feasibility gives

$$\begin{aligned} \boldsymbol{H}_{L_A}(\boldsymbol{x}^*, \boldsymbol{\lambda}^*; \rho) &= \boldsymbol{H}_L(\boldsymbol{x}^*, \boldsymbol{\lambda}^*) + \rho \boldsymbol{A}(\boldsymbol{x}^*)\boldsymbol{A}(\boldsymbol{x}^*)^{\mathsf{T}}, \\ \boldsymbol{d}^{\mathsf{T}} \boldsymbol{H}_{L_A}(\boldsymbol{x}^*, \boldsymbol{\lambda}^*; \rho)\boldsymbol{d} &= \boldsymbol{d}^{\mathsf{T}} \boldsymbol{H}_L(\boldsymbol{x}^*, \boldsymbol{\lambda}^*)\boldsymbol{d} + \rho \left\| \boldsymbol{A}(\boldsymbol{x}^*)^{\mathsf{T}} \boldsymbol{d} \right\|^2. \end{aligned} \tag{15.14}$$

On the tangent space $\boldsymbol{A}(\boldsymbol{x}^*)^{\mathsf{T}}\boldsymbol{d} = \boldsymbol{0}$, the penalty adds no curvature; positivity there must come from SOSC. In other directions, the Lagrangian curvature can be negative, but the penalty contributes a positive term. Under LICQ and SOSC, a sufficiently large finite ρ makes the whole Hessian positive definite. A positive penalty of arbitrary size cannot ensure this in general, as the saddle in the preceding example shows. Some problems, including the preceding convex quadratic benchmark, already have suitable curvature for every $\rho > 0$; the following theorem gives a general sufficient guarantee.

Theorem 15.6 (Finite-parameter local exactness at the correct multiplier)**.** *Suppose* $(\boldsymbol{x}^*, \boldsymbol{\lambda}^*)$ *is a feasible KKT pair, LICQ holds, and*

$$\boldsymbol{d}^{\mathsf{T}} \boldsymbol{H}_L(\boldsymbol{x}^*, \boldsymbol{\lambda}^*)\boldsymbol{d} > 0 \quad \textit{for every } \boldsymbol{d} \neq \boldsymbol{0} \textit{ with } \boldsymbol{A}(\boldsymbol{x}^*)^{\mathsf{T}}\boldsymbol{d} = \boldsymbol{0}.$$

For all sufficiently large finite ρ, $\boldsymbol{x}^*$ *is a strict local minimizer of* $L_A(\cdot, \boldsymbol{\lambda}^*; \rho)$.

Proof. At $\boldsymbol{x}^*$, feasibility and stationarity give zero augmented gradient. The augmented Hessian is $\boldsymbol{H}_L(\boldsymbol{x}^*, \boldsymbol{\lambda}^*) + \rho \boldsymbol{A}(\boldsymbol{x}^*)\boldsymbol{A}(\boldsymbol{x}^*)^{\mathsf{T}}$. If it were not positive definite for all sufficiently large ρ, there would exist $\rho^{(j)} \to \infty$ and unit vectors $\boldsymbol{d}^{(j)}$ with nonpositive quadratic forms. A subsequence tends to a unit vector $\bar{\boldsymbol{d}}$. Boundedness of the Lagrangian-Hessian term implies $\left\| \boldsymbol{A}(\boldsymbol{x}^*)^{\mathsf{T}} \boldsymbol{d}^{(j)} \right\|^2 = O(1/\rho^{(j)})$, so $\boldsymbol{A}(\boldsymbol{x}^*)^{\mathsf{T}}\bar{\boldsymbol{d}} = \boldsymbol{0}$. Dropping the nonnegative penalty term and

passing to the limit gives $\bar{\boldsymbol{d}}^{\mathsf{T}} \boldsymbol{H}_L(\boldsymbol{x}^*, \boldsymbol{\lambda}^*)\bar{\boldsymbol{d}} \leq 0$, contradicting SOSC. The unconstrained SOSC proves strict local minimality. □

For convex objectives and affine equalities, the same question has a global answer. The supporting-hyperplane inequality for the ordinary Lagrangian provides the required comparison.

Proposition 15.7 (Global exactness for convex affine equality problems)**.** *Suppose f is convex on an open convex $\mathcal{D}$, $\boldsymbol{g}$ is affine, and $(\boldsymbol{x}^*, \boldsymbol{\lambda}^*)$ is a feasible KKT pair. For every $\rho > 0$, $\boldsymbol{x}^*$ globally minimizes $L_A(\cdot, \boldsymbol{\lambda}^*; \rho)$. Every global minimizer of this augmented function is feasible and globally minimizes the original problem.*

Proof. The function $L(\cdot, \boldsymbol{\lambda}^*)$ is convex and has zero gradient at $\boldsymbol{x}^*$. Thus it is globally minimized there, and

$$L_A(\boldsymbol{x}, \boldsymbol{\lambda}^*; \rho) \geq f(\boldsymbol{x}^*) + \frac{\rho}{2} \|\boldsymbol{g}(\boldsymbol{x})\|^2 \geq f(\boldsymbol{x}^*) = L_A(\boldsymbol{x}^*, \boldsymbol{\lambda}^*; \rho).$$

Equality requires $\boldsymbol{g}(\boldsymbol{x}) = \boldsymbol{0}$ and $f(\boldsymbol{x}) = f(\boldsymbol{x}^*)$. □

To see why the convex assumptions matter, consider $f(x) = -x^4$, $g(x) = x$. The feasible KKT pair $(0, 0)$ satisfies LICQ, and equality SOSC holds vacuously because the tangent space is $\{0\}$. For every finite $\rho > 0$, $L_A(x, 0; \rho) = -x^4 + \rho x^2/2$ has a strict local minimum at zero but is unbounded below. Thus the local theorem does not imply a global augmented minimizer in a nonconvex problem.

15.4.3 The classic outer iteration

We can now combine the inner gradient test with the correction Equation (15.10). The multiplier estimate changes after each completed inner solve, while the penalty and inner tolerance stay fixed.

Algorithm: Classic ALM

Choose $\boldsymbol{x}^{(0)}$, $\boldsymbol{\lambda}^{(0)}$, a fixed penalty $\rho^{(k)} \equiv \rho^{(0)} > 0$, a fixed inner tolerance $0 < \varepsilon_{\text{in}}^{(k)} \equiv \varepsilon_{\text{in}}^{(0)} < \varepsilon_{\text{out}}$, and outer KKT tolerance $\varepsilon_{\text{out}} > 0$. At iteration k:

1. Warm-start and approximately minimize $L_A(\cdot, \boldsymbol{\lambda}^{(k)}; \rho^{(0)})$ until
$$\left\| \nabla_{\boldsymbol{x}} L_A(\boldsymbol{x}^{(k+1)}, \boldsymbol{\lambda}^{(k)}; \rho^{(0)}) \right\|_\infty \leq \varepsilon_{\text{in}}^{(k)}.$$
2. Unconditionally update $\boldsymbol{\lambda}^{(k+1)} = \boldsymbol{\lambda}^{(k)} + \rho^{(0)} \boldsymbol{g}(\boldsymbol{x}^{(k+1)})$, and compute $r_{\text{stat}}(\boldsymbol{x}^{(k+1)}, \boldsymbol{\lambda}^{(k+1)})$ and $r_{\text{feas}}(\boldsymbol{x}^{(k+1)})$.
3. Stop if their maximum is at most ε_{out}. Otherwise continue with the same penalty and inner tolerance.

The implementation uses the common inner solvers and failure handling described in Section 15.3.4. The numerical choices appear in Section 15.5.

15.4.4 Infeasible local branches

To test whether multiplier corrections alone rule out an infeasible local branch, return to Example 15.2 and examine the augmented subproblem at the same point.

Example 15.4 (Exact local ALM inner minima can remain infeasible)**.** At $x = 1$ in Example 15.2,

$$L'_A(1, \lambda; \rho) = 0, \qquad L''_A(1, \lambda; \rho) = 2 + 6(\lambda + \rho).$$

For $\lambda \geq 0$, this is an infeasible strict local minimum. Start from $x^{(0)} = 1$ and $\lambda^{(0)} = 0$. Warm-started exact local inner solves can keep returning $x^{(k+1)} = 1$, while the classic unconditional update gives

$$\lambda^{(k+1)} = \lambda^{(k)} + \rho^{(0)} = (k + 1)\rho^{(0)} \geq 0.$$

The next augmented subproblem still has a strict local minimum at 1, for the chosen fixed positive penalty. The updated-multiplier stationarity residual is zero, but the feasibility violation remains one. Thus the traditional unconditional update does not repair this infeasible local branch.

15.5 QP versus ALM: fixed accuracy and practical choices

15.5.1 A comparison with the same inner tolerance

To compare the work required by the two traditional methods, apply them to the same quadratic test with $\boldsymbol{x}^{(0)} = (2, 1)^\mathsf{T}$, $\lambda^{(0)} = 0$, and Armijo backtracking. Both algorithms and both subsolvers use the same fixed inner gradient tolerance $\varepsilon_{\text{in}}^{(k)} \equiv 10^{-7}$, with outer tolerance $\varepsilon_{\text{out}} = 10^{-6}$ and no inner iteration limit. QP starts at $\rho^{(0)} = 1$ and grows the penalty by 5; ALM keeps $\rho^{(k)} \equiv 10$ and uses Equation (15.10). We record both equality residuals after every completed inner solve.

To interpret the plotted quantity, apply the gradient identities derived for the two methods. Every completed inner solve has $r_{\text{stat}} \leq 10^{-7}$, using the QP multiplier estimate or ALM's updated multiplier. In the recorded comparison, stationarity is also no larger than feasibility at each point. Thus the plotted r_{KKT} equals r_{feas} on these histories, and the remaining outer work is to reduce feasibility below 10^{-6}. This is a verified feature of this experiment, not a general license to omit stationarity from stopping checks.

Method/subsolver	Outer	Total inner	Final ρ	r_{KKT}
QP/SD	10	3,836,166	1953125	8.192×10^{-7}
QP/Newton	10	10	1953125	8.192×10^{-7}
ALM/SD	6	277	10	3.393×10^{-7}
ALM/Newton	6	6	10	3.304×10^{-7}

All four runs reach the requested target. The large QP/SD count records actual accumulated work. To explain the contrast with ALM, compare the subproblem Hessians for affine equality constraints at the same penalty:

$$\boldsymbol{H}_{L_A}(\boldsymbol{x}, \boldsymbol{\lambda}; \rho) = \boldsymbol{H}_{P_\rho}(\boldsymbol{x}) = \boldsymbol{H}_f(\boldsymbol{x}) + \rho \boldsymbol{A}\boldsymbol{A}^\mathsf{T}.$$

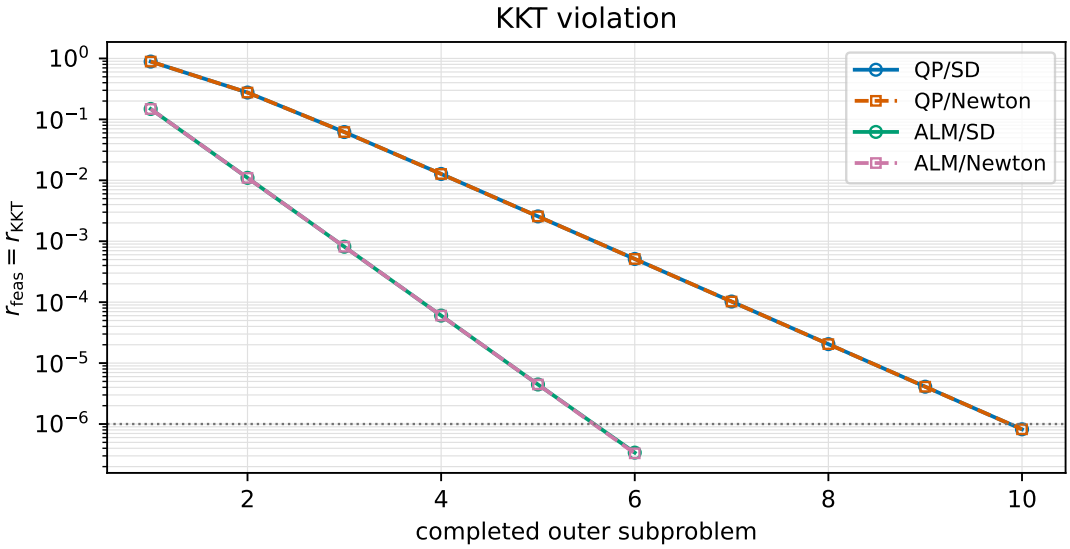


Figure 15.5: QP and ALM with fixed inner gradient tolerance $\varepsilon_{\text{in}}^{(k)} \equiv 10^{-7}$ and outer KKT tolerance $\varepsilon_{\text{out}} = 10^{-6}$. The curves show feasibility, which equals the recorded KKT violation on these histories.

Thus the multiplier term does not itself improve conditioning in this case. ALM's advantage in this experiment is that its multiplier corrections permit the target to be reached with $\rho = 10$, while QP uses much larger late penalties. This is not a universal claim that ALM is faster or better conditioned: total work also depends on the initialization, inner solver, accuracy target, and fixed penalty or growth-factor choice. Moreover, a Newton step solves a linear system, so iteration counts alone do not measure total computational cost.

15.5.2 Coordinate outer accuracy, inner accuracy, and penalty growth

The experiments suggest choosing the three accuracies and scales together: the required outer residual, the fixed inner gradient tolerance, and the penalty parameters. For the reported finite target, the inner tolerance is smaller than the outer one. Exact convergence as the iteration continues indefinitely is a different question, addressed by the error assumptions in Section 15.7.

To examine the penalty choice in ALM more closely, suppose an affine-constraint subproblem returns $\widetilde{\boldsymbol{x}} = \boldsymbol{x}_{\text{exact}} + \boldsymbol{e}$. Then $\boldsymbol{g}(\widetilde{\boldsymbol{x}}) = \boldsymbol{g}(\boldsymbol{x}_{\text{exact}}) + \boldsymbol{A}^{\mathsf{T}}\boldsymbol{e}$, and substitution in the multiplier update gives

$$\widetilde{\boldsymbol{\lambda}}^{+} = \boldsymbol{\lambda}_{\text{exact}}^{+} + \rho \boldsymbol{A}^{\mathsf{T}}\boldsymbol{e}.$$

A larger penalty can therefore amplify a point or constraint-evaluation error. This does not mean that a fixed gradient tolerance is amplified by exactly the same factor: the larger Hessian may reduce the associated point error. For the convex quadratic benchmark used in the QP–ALM comparison, the exact recurrence derived earlier contracts faster as ρ increases. The practical choice balances this improvement against the conditioning and inner work seen in the growth-factor experiment.

15.6 Extensions to inequality constraints

To extend the construction to inequalities, first identify which part of a constraint should be penalized. For $h_j(\boldsymbol{x}) \leq 0$, only its positive part represents a violation. This gives the mixed QP model

$$P_\rho(\boldsymbol{x}) = f(\boldsymbol{x}) + \frac{\rho}{2}\left(\|\boldsymbol{g}(\boldsymbol{x})\|^2 + \|[\boldsymbol{h}(\boldsymbol{x})]_+\|^2\right).$$

Differentiating $[h]_+^2/2$ with respect to h gives $[h]_+$, including at zero. The inequality term in its gradient therefore has coefficient $\widehat{\boldsymbol{\nu}} = \rho[\boldsymbol{h}(\boldsymbol{x})]_+ \geq \boldsymbol{0}$.

For ALM, derive the inequality term by eliminating an auxiliary variable. A scalar inequality $h \leq 0$ can be written $h - s = 0$, $s \leq 0$. At a current multiplier $\nu \geq 0$, the equality augmented term is $\nu(h-s) + \rho(h-s)^2/2$. Complete the square:

$$\nu(h-s) + \frac{\rho}{2}(h-s)^2 = \frac{\rho}{2}(s - h - \nu/\rho)^2 - \frac{\nu^2}{2\rho}.$$

Minimizing over $s \leq 0$ projects $h + \nu/\rho$ onto that half-line, so $s^* = \min\{0, h + \nu/\rho\}$. Substitution yields

$$\psi(h, \nu; \rho) = \min_{s \leq 0}\left\{\nu(h-s) + \frac{\rho}{2}(h-s)^2\right\} = \frac{[\max\{0, \nu + \rho h\}]^2 - \nu^2}{2\rho}.$$

Below the join $\nu + \rho h = 0$, this expression is constant in h; above it, differentiation gives $\nu + \rho h$. Both one-sided derivatives are zero at the join, hence

$$\frac{\partial \psi}{\partial h} = \max\{0, \nu + \rho h\}$$

is continuous. The second derivatives are 0 and ρ on the two sides, so ψ is not twice differentiable at the join.

Adding the eliminated terms for all inequalities gives the Powell–Hestenes–Rockafellar augmented Lagrangian

$$L_A(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{\nu}; \rho) = f(\boldsymbol{x}) + \boldsymbol{\lambda}^\mathsf{T}\boldsymbol{g}(\boldsymbol{x}) + \frac{\rho}{2}\|\boldsymbol{g}(\boldsymbol{x})\|^2 + \frac{\|[\boldsymbol{\nu} + \rho\boldsymbol{h}(\boldsymbol{x})]_+\|^2 - \|\boldsymbol{\nu}\|^2}{2\rho}.$$

Its gradient identifies the classical multiplier corrections:

$$\boldsymbol{\lambda}^+ = \boldsymbol{\lambda} + \rho\boldsymbol{g}(\boldsymbol{x}), \qquad \boldsymbol{\nu}^+ = [\boldsymbol{\nu} + \rho\boldsymbol{h}(\boldsymbol{x})]_+.$$

Projection preserves dual feasibility, while the other KKT components, including complementarity, still enter the general stopping test. The piecewise second derivative also explains why a smooth Newton treatment must account for switching points.

Simply adding $\boldsymbol{\nu}^\mathsf{T}\boldsymbol{h} + \rho\|[\boldsymbol{h}]_+\|^2/2$ would not produce this model. For an inactive inequality with $h < 0$, its coefficient is ν, whereas the eliminated model uses $\max\{0, \nu + \rho h\}$. The elimination is therefore needed to obtain the correct inequality multiplier correction.

15.7 Supplementary theory and unimplemented variants

We now examine the hypotheses behind the convergence results and the reasons one might consider changing the traditional algorithms. All adaptive rules, accuracy schedules, and deliberately truncated inner solves are confined to this section as mathematical reference material. None is implemented or used in the numerical results of this chapter. The main comparison remains the fixed-accuracy, fixed-rule experiment in Section 15.5.

15.7.1 QP: approximately global inner solves

First return to Equation (15.7) and allow a nonzero error in the global objective comparison. Keeping that error in the same inequality shows which conclusions require it to vanish. This is an objective-value accuracy assumption, not the gradient stopping criterion used by the implemented subsolver.

Theorem 15.8 (Approximately global inner solves)**.** *Under the assumptions of Theorem 15.2, replace exact global minimization by the following inner objective-value tolerance condition:*

$$P_{\rho^{(k)}}(\boldsymbol{x}^{(k+1)}) \le \inf_{\boldsymbol{x}\in\mathcal{D}} P_{\rho^{(k)}}(\boldsymbol{x}) + \varepsilon_{\text{in}}^{(k)}, \qquad \varepsilon_{\text{in}}^{(k)} \ge 0, \quad \varepsilon_{\text{in}}^{(k)} \to 0.$$

Then every cluster point in $\mathcal{D}$ is globally optimal. If f is bounded below along the returned points and $\varepsilon_{\text{in}}^{(k)}/\rho^{(k)} \to 0$, their feasibility violations tend to zero, even without $\varepsilon_{\text{in}}^{(k)} \to 0$.

Proof. Comparison with a feasible $\boldsymbol{y}$ yields

$$f(\boldsymbol{x}^{(k+1)}) + \frac{\rho^{(k)}}{2}\left\|\boldsymbol{g}(\boldsymbol{x}^{(k+1)})\right\|^2 \le f(\boldsymbol{y}) + \varepsilon_{\text{in}}^{(k)}.$$

For an interior convergent subsequence, continuity bounds the objective values and rearrangement gives

$$\left\|\boldsymbol{g}(\boldsymbol{x}^{(k_j+1)})\right\|^2 \le \frac{2[f(\boldsymbol{y}) - f(\boldsymbol{x}^{(k_j+1)}) + \varepsilon_{\text{in}}^{(k_j)}]}{\rho^{(k_j)}} \longrightarrow 0.$$

The limit is therefore feasible. Dropping the nonnegative penalty term also gives $f(\boldsymbol{x}^{(k_j+1)}) \le f(\boldsymbol{y}) + \varepsilon_{\text{in}}^{(k_j)}$. Taking limits and using the arbitrariness of feasible $\boldsymbol{y}$ proves optimality. For the second assertion, the lower bound $f(\boldsymbol{x}^{(k+1)}) \ge a$ gives

$$\left\|\boldsymbol{g}(\boldsymbol{x}^{(k+1)})\right\|^2 \le \frac{2[f(\boldsymbol{y}) - a]}{\rho^{(k)}} + \frac{2\varepsilon_{\text{in}}^{(k)}}{\rho^{(k)}} \longrightarrow 0.$$

□

15.7.2 QP: stationary inner solves and their limits

The global QP analysis used objective comparisons with every feasible point. We now ask what remains when only inner gradients tend to zero. Divide the inner stationarity equation by the growing penalty, then use the constraint Jacobian to recover feasibility.

Theorem 15.9 (KKT limit of inexact stationary penalty points)**.** *Suppose f, g_i are continuously differentiable, $\rho^{(k)} \to \infty$, $\boldsymbol{x}^{(k+1)} \to \boldsymbol{x}^* \in \mathcal{D}$, and $\left\|\nabla P_{\rho^{(k)}}(\boldsymbol{x}^{(k+1)})\right\| \to 0$. If $\boldsymbol{A}(\boldsymbol{x}^*)$ has full column rank, then $\boldsymbol{x}^*$ is feasible and $\widehat{\boldsymbol{\lambda}}^{(k+1)} = \rho^{(k)}\boldsymbol{g}(\boldsymbol{x}^{(k+1)})$ converges to its unique KKT multiplier.*

Proof. Set $\boldsymbol{e}^{(k+1)} = \nabla P_{\rho^{(k)}}(\boldsymbol{x}^{(k+1)})$. Then

$$\boldsymbol{A}(\boldsymbol{x}^{(k+1)})\boldsymbol{g}(\boldsymbol{x}^{(k+1)}) = \frac{\boldsymbol{e}^{(k+1)} - \nabla f(\boldsymbol{x}^{(k+1)})}{\rho^{(k)}} \to \boldsymbol{0}.$$

Continuity gives $\boldsymbol{A}(\boldsymbol{x}^*)\boldsymbol{g}(\boldsymbol{x}^*) = \boldsymbol{0}$, so full column rank gives feasibility. For large k, the Gram matrix is nonsingular and

$$\widehat{\boldsymbol{\lambda}}^{(k+1)} = [\boldsymbol{A}(\boldsymbol{x}^{(k+1)})^{\mathsf{T}}\boldsymbol{A}(\boldsymbol{x}^{(k+1)})]^{-1}\boldsymbol{A}(\boldsymbol{x}^{(k+1)})^{\mathsf{T}}[\boldsymbol{e}^{(k+1)} - \nabla f(\boldsymbol{x}^{(k+1)})].$$

Its right-hand side converges. The limiting gradient equation gives stationarity, and full column rank makes the multiplier unique. □

This gives a KKT limit, while Theorem 15.8 gives a global optimum under approximately global solves. Full rank plays a second role here: it makes the limiting multiplier unique. To separate uniqueness from convergence of an estimate, consider redundant equalities.

Example 15.5 (Multiplier estimates without multiplier uniqueness)**.** For $\min_x x$ subject to $g_1(x) = x = 0$, $g_2(x) = 2x = 0$, form the penalty $P_\rho(x) = x + (5\rho/2)x^2$. Its derivative is $1 + 5\rho x$, so the unique inner minimizer is $x(\rho) = -1/(5\rho)$. Substituting it into the estimate gives

$$\rho(g_1(x(\rho)), g_2(x(\rho)))^{\mathsf{T}} = (-1/5, -2/5)^{\mathsf{T}}.$$

This estimate converges, but the KKT stationarity equation $1 + \lambda_1 + 2\lambda_2 = 0$ admits infinitely many multipliers. Thus convergence of an estimate alone does not establish uniqueness.

15.7.3 ALM: local convergence with a fixed finite penalty

The finite-exactness result fixes the multiplier at its unknown correct value. To justify computing that multiplier, we next study the map from a nearby current estimate to the next corrected estimate. The implicit-function theorem will make this map precise and supply a contraction estimate.

Theorem 15.10 (Local exact ALM multiplier contraction)**.** *Under the assumptions of Theorem 15.6, there are neighborhoods of $\boldsymbol{x}^*, \boldsymbol{\lambda}^*$, constants $\rho_0, M > 0$, and a local minimizing branch $\boldsymbol{x}(\rho, \boldsymbol{\lambda})$, such that, for $\rho \geq \rho_0$,*

$$\|\boldsymbol{x}(\rho, \boldsymbol{\lambda}) - \boldsymbol{x}^*\| \leq \frac{M}{\rho}\|\boldsymbol{\lambda} - \boldsymbol{\lambda}^*\|,$$
$$\left\|\boldsymbol{\lambda}^+(\rho, \boldsymbol{\lambda}) - \boldsymbol{\lambda}^*\right\| \leq \frac{M}{\rho}\|\boldsymbol{\lambda} - \boldsymbol{\lambda}^*\|, \qquad \boldsymbol{\lambda}^+ = \boldsymbol{\lambda} + \rho\boldsymbol{g}(\boldsymbol{x}(\rho, \boldsymbol{\lambda})).$$

A sufficiently large fixed finite penalty, sufficiently close initial multiplier, and exact local inner solves on this branch give convergence of both the multipliers and returned points to the KKT pair.

Proof. For this proof only, use the auxiliary variable $\varepsilon = 1/\rho$. Inner stationarity and the multiplier correction are equivalent to

$$\mathcal{F}(\boldsymbol{x}, \boldsymbol{\lambda}^+; \varepsilon, \boldsymbol{\lambda}) = \begin{pmatrix} \nabla f(\boldsymbol{x}) + \boldsymbol{A}(\boldsymbol{x})\boldsymbol{\lambda}^+ \\ \boldsymbol{g}(\boldsymbol{x}) - \varepsilon(\boldsymbol{\lambda}^+ - \boldsymbol{\lambda}) \end{pmatrix} = \boldsymbol{0}. \tag{15.15}$$

At $(\boldsymbol{x}^*, \boldsymbol{\lambda}^*; 0, \boldsymbol{\lambda}^*)$, the derivative with respect to $(\boldsymbol{x}, \boldsymbol{\lambda}^+)$ is

$$\boldsymbol{K} = \begin{pmatrix} \boldsymbol{H}_L(\boldsymbol{x}^*, \boldsymbol{\lambda}^*) & \boldsymbol{A}(\boldsymbol{x}^*) \\ \boldsymbol{A}(\boldsymbol{x}^*)^{\mathsf{T}} & \boldsymbol{0} \end{pmatrix}.$$

To check nonsingularity, suppose $\boldsymbol{H}_L\boldsymbol{d}+\boldsymbol{A}\boldsymbol{y}=\boldsymbol{0}$, $\boldsymbol{A}^{\mathsf{T}}\boldsymbol{d}=\boldsymbol{0}$. Multiplication by $\boldsymbol{d}^{\mathsf{T}}$ and SOSC force $\boldsymbol{d}=\boldsymbol{0}$. LICQ then forces $\boldsymbol{y}=\boldsymbol{0}$.

The implicit-function theorem supplies unique continuously differentiable local functions $\boldsymbol{x}(\varepsilon,\boldsymbol{\lambda})$ and $\boldsymbol{\lambda}^+(\varepsilon,\boldsymbol{\lambda})$. After shrinking neighborhoods, the inverse Jacobian with respect to the unknowns is uniformly bounded. Since

$$D_{\boldsymbol{\lambda}}\mathcal{F}=\begin{pmatrix}\boldsymbol{0}\\ \varepsilon\boldsymbol{I}\end{pmatrix},$$

implicit differentiation gives $\|D_{\boldsymbol{\lambda}}\boldsymbol{x}\|+\|D_{\boldsymbol{\lambda}}\boldsymbol{\lambda}^+\|\le M\varepsilon$. At $\boldsymbol{\lambda}=\boldsymbol{\lambda}^*$, the branch is exactly $\boldsymbol{x}(\varepsilon,\boldsymbol{\lambda}^*)=\boldsymbol{x}^*$ and $\boldsymbol{\lambda}^+(\varepsilon,\boldsymbol{\lambda}^*)=\boldsymbol{\lambda}^*$. Integrate the derivative bound along the multiplier line segment to obtain the two estimates.

It remains to verify that the branch is minimizing. LICQ and strict positivity of the Lagrangian Hessian on the tangent space persist uniformly in a sufficiently small neighborhood of the KKT pair. Otherwise a sequence of nearby pairs and unit tangent vectors would converge to a unit tangent vector with a nonpositive quadratic form at the KKT pair, contradicting SOSC. On a compact smaller neighborhood, the argument from Theorem 15.6 is uniform: if no common sufficiently large ρ made $\boldsymbol{H}_L(\boldsymbol{x},\boldsymbol{\lambda}^+)+\rho\boldsymbol{A}(\boldsymbol{x})\boldsymbol{A}(\boldsymbol{x})^{\mathsf{T}}$ positive definite, a convergent subsequence of violating unit vectors would yield a nonzero tangent limit contradicting the uniform strict positivity. Thus the stationary branch consists of strict local minima for all sufficiently large ρ.

Choose a fixed ρ with $M/\rho<1$ and a closed multiplier ball contained in the branch neighborhood. The multiplier update maps this ball into itself and contracts the distance to $\boldsymbol{\lambda}^*$. Iterating the estimate proves multiplier convergence; the first estimate gives primal convergence. □

To include inner errors, allow a nonzero residual in the same local system. The derivative with respect to that residual adds one term to the estimate.

Proposition 15.11 (A local estimate for inexact ALM corrections)**.** *In a neighborhood of the branch from Theorem 15.10, suppose the inner gradient residual is $\boldsymbol{e}$ and the updated multiplier is $\boldsymbol{\lambda}^+=\boldsymbol{\lambda}+\rho\boldsymbol{g}(\boldsymbol{x})$. For nearby points and sufficiently small residuals, some constant M_1, independent of sufficiently large ρ, gives*

$$\|\boldsymbol{x}-\boldsymbol{x}^*\|+\left\|\boldsymbol{\lambda}^+-\boldsymbol{\lambda}^*\right\|\le M_1\left(\frac{\|\boldsymbol{\lambda}-\boldsymbol{\lambda}^*\|}{\rho}+\|\boldsymbol{e}\|\right).$$

Proof. Replace the first component of Equation (15.15) by $\nabla f+\boldsymbol{A}\boldsymbol{\lambda}^+-\boldsymbol{e}$. The same nonsingular Jacobian gives a local solution map in $\boldsymbol{e}$. Its derivative with respect to $\boldsymbol{e}$ is uniformly bounded, whereas its multiplier derivative is $O(1/\rho)$. Integrating along a line segment from $(\boldsymbol{\lambda}^*,\boldsymbol{0})$ to $(\boldsymbol{\lambda},\boldsymbol{e})$ proves the estimate. □

The bound separates the contracted multiplier error from the inner error. It applies to the stated local branch; an arbitrary inner solver need not select that branch. We will examine a concrete fixed-error sequence below.

15.7.4 ALM: a global convex-affine guarantee

For a convex objective and affine equalities, we can compare iterates with a KKT pair globally. The main step is to combine gradient monotonicity with the multiplier correction and expand the squared multiplier distance.

Theorem 15.12 (Fixed-penalty exact ALM for a convex affine problem)**.** *Suppose $\mathcal{D}$ is open and convex, f is continuously differentiable and convex, $\boldsymbol{g}$ is affine, and a feasible KKT pair $(\boldsymbol{x}^*, \boldsymbol{\lambda}^*)$ exists. Fix $\rho > 0$. At each iteration, globally minimize $L_A(\cdot, \boldsymbol{\lambda}^{(k)}; \rho)$ and perform the unconditional update $\boldsymbol{\lambda}^{(k+1)} = \boldsymbol{\lambda}^{(k)} + \rho \boldsymbol{g}(\boldsymbol{x}^{(k+1)})$. Then*

$$\left\|\boldsymbol{\lambda}^{(k+1)} - \boldsymbol{\lambda}^*\right\|^2 + \rho^2 \left\|\boldsymbol{g}(\boldsymbol{x}^{(k+1)})\right\|^2 \leq \left\|\boldsymbol{\lambda}^{(k)} - \boldsymbol{\lambda}^*\right\|^2. \tag{15.16}$$

Consequently the multipliers are bounded and feasibility violations tend to zero. If the primal points remain in a compact subset of $\mathcal{D}$, the multipliers converge and every primal cluster point is globally optimal.

Proof. An attained inner minimizer in an open domain has zero gradient, so $\nabla f(\boldsymbol{x}^{(k+1)}) + \boldsymbol{A}\boldsymbol{\lambda}^{(k+1)} = \boldsymbol{0}$. Write the two first-order convex inequalities

$$f(\boldsymbol{y}) \geq f(\boldsymbol{x}) + \nabla f(\boldsymbol{x})^\mathsf{T}(\boldsymbol{y} - \boldsymbol{x}), \qquad f(\boldsymbol{x}) \geq f(\boldsymbol{y}) + \nabla f(\boldsymbol{y})^\mathsf{T}(\boldsymbol{x} - \boldsymbol{y}).$$

Adding and rearranging gives gradient monotonicity:

$$(\boldsymbol{x} - \boldsymbol{y})^\mathsf{T}(\nabla f(\boldsymbol{x}) - \nabla f(\boldsymbol{y})) \geq 0.$$

Apply it to $\boldsymbol{x}^{(k+1)}, \boldsymbol{x}^*$, and use stationarity at both points. Since the equality is affine, $\boldsymbol{g}(\boldsymbol{x}^{(k+1)}) = \boldsymbol{A}^\mathsf{T}(\boldsymbol{x}^{(k+1)} - \boldsymbol{x}^*)$, and hence

$$[\boldsymbol{g}(\boldsymbol{x}^{(k+1)})]^\mathsf{T}(\boldsymbol{\lambda}^{(k+1)} - \boldsymbol{\lambda}^*) \leq 0.$$

Write $\boldsymbol{d}_\lambda = \boldsymbol{\lambda}^{(k+1)} - \boldsymbol{\lambda}^{(k)} = \rho \boldsymbol{g}(\boldsymbol{x}^{(k+1)})$. Expanding a squared norm gives

$$\begin{aligned}\left\|\boldsymbol{\lambda}^{(k)} - \boldsymbol{\lambda}^*\right\|^2 &= \left\|\boldsymbol{\lambda}^{(k+1)} - \boldsymbol{\lambda}^*\right\|^2 + \|\boldsymbol{d}_\lambda\|^2 - 2\boldsymbol{d}_\lambda^\mathsf{T}(\boldsymbol{\lambda}^{(k+1)} - \boldsymbol{\lambda}^*) \\ &\geq \left\|\boldsymbol{\lambda}^{(k+1)} - \boldsymbol{\lambda}^*\right\|^2 + \|\boldsymbol{d}_\lambda\|^2.\end{aligned}$$

This is the claimed inequality. Summing it yields $\sum_k \rho^2 \left\|\boldsymbol{g}(\boldsymbol{x}^{(k+1)})\right\|^2 < \infty$, so feasibility tends to zero and multipliers stay bounded.

Compact primal iterates and bounded multipliers give a joint cluster pair $(\bar{\boldsymbol{x}}, \bar{\boldsymbol{\lambda}})$. Feasibility and stationarity pass to the limit. It is KKT and therefore globally optimal by Theorem 14.1. The same inequality applies with $\bar{\boldsymbol{\lambda}}$ instead of $\boldsymbol{\lambda}^*$. Thus the distance to $\bar{\boldsymbol{\lambda}}$ is nonincreasing and has a subsequence tending to zero, so the whole multiplier sequence converges to it. Every primal cluster point then satisfies KKT with that multiplier and is globally optimal. □

To determine a sufficient accuracy condition for inexact inner solves, repeat the distance calculation with the residual retained. Summability will keep the accumulated perturbation finite.

Proposition 15.13 (Summable inner errors in the convex-affine case)**.** *Under the convex-affine assumptions above, keep primal points in a compact subset of $\mathcal{D}$, fix $\rho > 0$, and update the multipliers unconditionally by Equation (15.10). If inner gradient residuals satisfy $\sum_k \left\|\boldsymbol{e}^{(k+1)}\right\| < \infty$, feasibility tends to zero, multipliers converge, and all primal cluster points are globally optimal.*

Proof. Now $\nabla f(\boldsymbol{x}^{(k+1)}) + \boldsymbol{A}\boldsymbol{\lambda}^{(k+1)} = \boldsymbol{e}^{(k+1)}$. If D bounds $\left\|\boldsymbol{x}^{(k+1)} - \boldsymbol{x}^*\right\|$, gradient monotonicity gives

$$[\boldsymbol{g}(\boldsymbol{x}^{(k+1)})]^{\mathsf{T}}(\boldsymbol{\lambda}^{(k+1)} - \boldsymbol{\lambda}^*) \le D\left\|\boldsymbol{e}^{(k+1)}\right\|.$$

The previous squared-distance calculation becomes

$$a_{k+1} + \rho^2\left\|\boldsymbol{g}(\boldsymbol{x}^{(k+1)})\right\|^2 \le a_k + \epsilon_k, \qquad a_k = \left\|\boldsymbol{\lambda}^{(k)} - \boldsymbol{\lambda}^*\right\|^2, \quad \epsilon_k = 2\rho D\left\|\boldsymbol{e}^{(k+1)}\right\|.$$

The series $\sum_k \epsilon_k$ converges. Therefore $a_k + \sum_{j=k}^{\infty} \epsilon_j$ is nonincreasing and nonnegative, proving boundedness and convergence of a_k. Summation also gives square summability of feasibility violations. A joint cluster pair is KKT because residuals vanish. Apply the same argument to its multiplier: the distance to that multiplier has a limit and a subsequence tending to zero, proving convergence of all multipliers. Every primal cluster point is then KKT and globally optimal. □

15.7.5 What deliberately truncating an inner solve changes

To see why an outer iteration cannot simply ignore an unfinished inner solve, consider the following nonlinear equality analytically.

Example 15.6 (A nonlinear branch and a truncated-step variant)**.** Let

$$f(x) = \tfrac{1}{2}x^2, \qquad g(x) = 1 + xe^{-x} = 0.$$

First locate the feasible point. Since $g'(x) = (1-x)e^{-x} > 0$ for $x < 0$, with $g(x) \to -\infty$ as $x \to -\infty$ and $g(0) = 1$, there is exactly one negative root. For $x \ge 0$, $g(x) \ge 1$, so no further root exists. The unique feasible point is $x^* \approx -0.5671432904$. We compare the location of this feasible point with the direction of an initial local inner step from $x = 3$. Writing $a = e^{-3}$, differentiation gives

$$P'_\rho(3) = 3 - 2\rho a(1+3a), \qquad P''_\rho(3) = 1 + \rho(a + 7a^2).$$

For $\rho = 100$, the gradient is negative and the Hessian is positive, so the Newton direction $-P'_\rho(3)/P''_\rho(3)$ is positive. Any positive Armijo step in that direction stays in $x > 3$, where $g(x) > 1$, rather than moving toward the negative feasible point. The first ALM subproblem with $\lambda^{(0)} = 0$ is identical to this QP subproblem and has the same initial direction.

If an alternative outer procedure takes just one such inner step and continues regardless of its gradient, it no longer enforces the inner accuracy condition in either stated algorithm. Moreover, even fully solved local subproblems need not be global minimizers; the preceding direction calculation is not a proof that all later iterates follow a particular branch. It therefore explains two distinct issues: an uncertified truncated step and the selection of a nonglobal branch. Neither issue is repaired merely by calling the outer method QP or ALM.

15.7.6 Alternative accuracy schedules and adaptive penalty rules

We can now use the preceding error estimates to motivate alternative accuracy choices. To enforce a vanishing or summable residual in an asymptotic theorem, one may decrease the requested tolerance as the outer iteration proceeds. The schedule must bound the quantity appearing in that theorem: an inner gradient norm and an objective-value error are different quantities.

For QP, one possible finite-target schedule is

$$\varepsilon_{\text{in}}^{(k)} = \max\{\varepsilon_{\text{out}}, \min\{1, 1/\rho^{(k)}\}\}.$$

For ALM, another possible schedule starts from $\rho^{(0)} > 0$ and uses

$$\varepsilon_{\text{in}}^{(k)} = \max\{0.1\varepsilon_{\text{out}}, (\rho^{(0)})^{-1}0.1^{k}\}.$$

In each case, the floor prevents the tolerance from tending to zero. Thus these finite-target rules alone do not meet a theorem's vanishing-error hypothesis. An asymptotic accuracy schedule must instead satisfy the specific error condition in the convergence result being invoked.

A different modification responds to the observed feasibility reduction. For ALM, after an unsuccessful outer KKT check, one could choose constants $\theta \in (0,1)$ and $\gamma > 1$, and set

$$\rho^{(k+1)} = \begin{cases} \rho^{(k)}, & r_{\text{feas}}(\boldsymbol{x}^{(k+1)}) \le \theta r_{\text{feas}}(\boldsymbol{x}^{(k)}), \\ \gamma\rho^{(k)}, & \text{otherwise.} \end{cases}$$

It differs from both the fixed-penalty ALM implemented here and the accept/reject variant below. Residual-based choices of QP penalties are also possible, but then divergence of the penalty must be verified separately before applying a theorem that assumes $\rho^{(k)} \to \infty$.

15.7.7 Advanced reference: a two-tolerance accept/reject variant of ALM

The following distinct variant uses the two-tolerance structure of Nocedal and Wright, *Numerical Optimization*, second edition, Section 17.4, Algorithm 17.4. An inner stationarity tolerance $\varepsilon_{\text{in}}^{(k)}$ and a separate feasibility tolerance $\eta^{(k)}$ decide whether to accept a multiplier correction or strengthen the penalty. This is not a literal comparison of differently scaled stationarity and feasibility residuals.

Algorithm: Two-tolerance ALM

Choose $\boldsymbol{x}^{(0)}$, $\boldsymbol{\lambda}^{(0)}$, $\rho^{(0)} > 1$, and growth factor $\gamma > 1$. Initialize $\varepsilon_{\text{in}}^{(0)} = 1/\rho^{(0)}$ and $\eta^{(0)} = (\rho^{(0)})^{-0.1}$. At iteration k:

1. Warm-start and approximately minimize $L_A(\cdot, \boldsymbol{\lambda}^{(k)}; \rho^{(k)})$ until

$$\left\| \nabla_{\boldsymbol{x}} L_A(\boldsymbol{x}^{(k+1)}, \boldsymbol{\lambda}^{(k)}; \rho^{(k)}) \right\|_\infty \le \varepsilon_{\text{in}}^{(k)}.$$

2. Form the trial multiplier $\boldsymbol{\lambda}_{\text{trial}}^{(k+1)} = \boldsymbol{\lambda}^{(k)} + \rho^{(k)}\boldsymbol{g}(\boldsymbol{x}^{(k+1)})$. Report stationarity with this trial estimate and compute feasibility. If $\max\{r_{\text{stat}}(\boldsymbol{x}^{(k+1)}, \boldsymbol{\lambda}_{\text{trial}}^{(k+1)}), r_{\text{feas}}(\boldsymbol{x}^{(k+1)})\} \le \varepsilon_{\text{out}}$, return the point and trial multiplier.

3. If $r_{\text{feas}} \le \eta^{(k)}$, accept

$$\boldsymbol{\lambda}^{(k+1)} = \boldsymbol{\lambda}_{\text{trial}}^{(k+1)}, \qquad \rho^{(k+1)} = \rho^{(k)},$$
$$\varepsilon_{\text{in}}^{(k+1)} = \varepsilon_{\text{in}}^{(k)}/\rho^{(k)}, \quad \eta^{(k+1)} = \eta^{(k)}/(\rho^{(k)})^{0.9}.$$

Otherwise retain $\boldsymbol{\lambda}^{(k+1)} = \boldsymbol{\lambda}^{(k)}$, set $\rho^{(k+1)} = \gamma\rho^{(k)}$, and reset

$$\varepsilon_{\text{in}}^{(k+1)} = 1/\rho^{(k+1)}, \qquad \eta^{(k+1)} = (\rho^{(k+1)})^{-0.1}.$$

Warning

For the advanced accept/reject rule above, after rejection the accepted multiplier and trial multiplier differ. The transformed gradient equals stationarity with the trial multiplier, not with the unchanged accepted estimate. A recorded small transformed gradient must be labeled accordingly.

15.8 Theory and implementation summary

The conclusions depend on the inner guarantee and problem assumptions. The following table collects those dependencies for comparison.

Setting	Assumptions	Conclusion
QP global path	Nonempty feasible set, global inner solves, $\rho \to \infty$, continuity, interior cluster points	Such cluster points are globally optimal; compactness ensures existence.
QP stationary path	$\rho \to \infty$, vanishing inner gradients, convergent points with limit in $\mathcal{D}$, full-rank constraint Jacobian at the limit	A feasible KKT limit with a unique multiplier, not a general minimum certificate.
ALM local path	LICQ, SOSC, nearby initialization, large fixed finite penalty, exact local inner solves on the correct branch	Local convergence to the KKT pair; controlled inner errors add corresponding error terms.
ALM convex affine path	A KKT pair exists, fixed positive penalty, global inner solutions; compact primal iterates	Feasibility tends to zero, multipliers converge, and primal cluster points are globally optimal.

For an implementation, first identify which inner guarantee is actually available, then use the matching row of the table to interpret its output. The computations in this chapter use finite gradient tolerances; their reported success means that the measured outer KKT test was met, rather than that a global inner minimum or an asymptotic theorem was verified numerically.

15.9 Exercises

Exercise 15.1 (Penalty path and nonexactness)**.** For $\min_x (x-3)^2$ subject to $2x-1=0$, derive the exact penalty minimizer, its feasibility violation, and multiplier estimate. Determine their rates as $\rho \to \infty$, and decide whether any finite penalty gives the constrained minimizer.

Exercise 15.2 (An incomplete stopping test)**.** For $\min_{\boldsymbol{x}\in\mathbb{R}^2} \frac{1}{2}[(x_1-1)^2 + x_2^2]$ subject to $x_2 = 0$, construct a feasible sequence whose stationarity violation stays bounded away from zero for every multiplier choice. Explain why feasibility alone cannot terminate either method.

Exercise 15.3 (Exact ALM correction)**.** For

$$\min_{\boldsymbol{x}\in\mathbb{R}^2} \frac{1}{2}(x_1^2 + 2x_2^2) \qquad \text{subject to} \qquad x_1 - x_2 - 2 = 0,$$

derive the exact multiplier recurrence at fixed $\rho > 0$. Compute its fixed point and contraction factor; compare it with the QP multiplier estimate.

Exercise 15.4 (Constraint scaling and conditioning)**.** For $\min_{\boldsymbol{x}\in\mathbb{R}^2} \frac{1}{2}(x_1^2 + 9x_2^2)$ subject to $a(x_1 + 2x_2 - 3) = 0$, $a > 0$, compute the penalty Hessian and eigenvalues. Explain how scaling a changes the effective penalty and multiplier, although not the constrained minimizer. Analyze the large-ρ tangent and normal curvature scales.

Exercise 15.5 (Inexact global solutions and compactness)**.** Adapt the compactness corollary to approximately global inner solutions with $\varepsilon_{\text{in}}^{(k)} \to 0$: construct one fixed compact sublevel set containing all sufficiently late returned points. State precisely the additional sublevel-set assumption needed.